

INTRODUCTION TO DEEP LEARNING

Course number: 00240332

Lecture 3: Regression and Classification

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Last lecture review

I. Math basics

PMF

PDF

Joint Prob

Marginal prob

Conditional prob

Gradient decent



II. Machine learning basics

– Task T

– Performance P

– Experience E

Model capacity

MLE

$$\theta_{\text{ML}} = \arg \max_{\theta} p_{\text{model}}(\mathbb{X}; \theta)$$

SGD

$(x^{(i)}, y^{(i)})$

The entire training set



Outline

- I. Logistic regression
- II. Softmax regression
- III. Summary

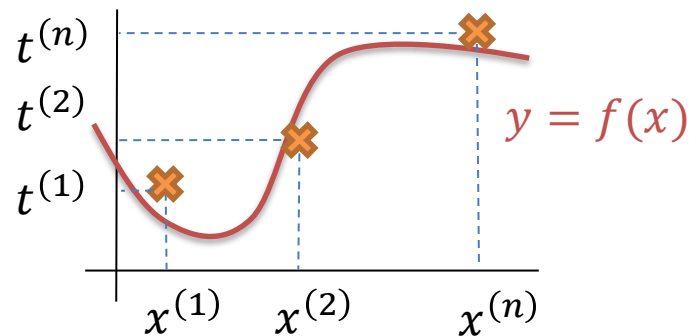
Regression and classification

Given a set of data points $\mathbf{x}^{(n)} \in \mathbb{R}^m$ and the corresponding labels $t^{(n)} \in \Omega$: $\{(\mathbf{x}^{(1)}, t^{(1)}), (\mathbf{x}^{(2)}, t^{(2)}), \dots, (\mathbf{x}^{(N)}, t^{(N)})\}$, for a new data point $\mathbf{x}$, predict its label

The goal is to find a mapping $f: \mathbb{R}^m \rightarrow \Omega$

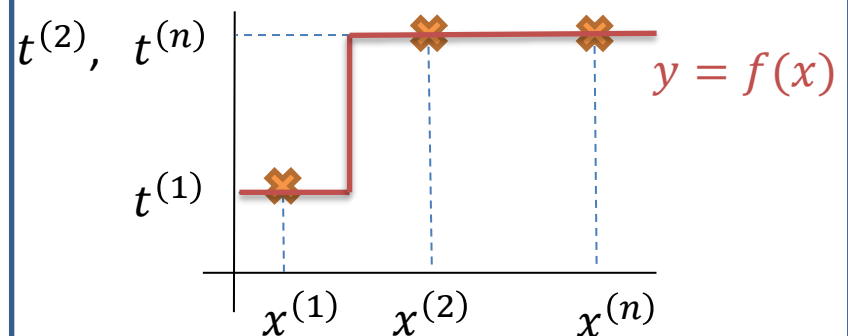
Regression

Ω is continuous



Classification

Ω is discrete



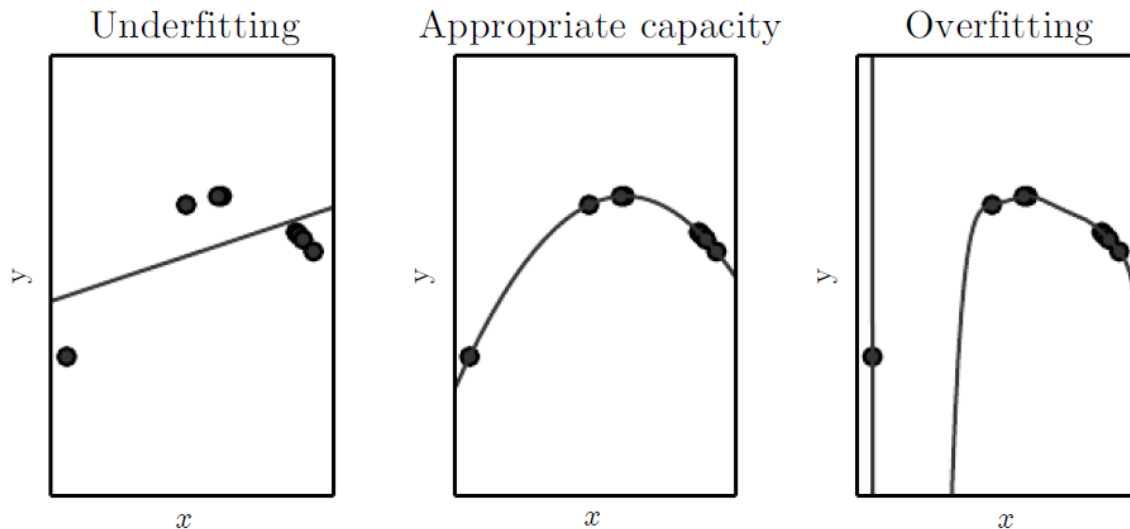
Recall: polynomial regression

- Consider a regression problem in which the input x and output y are both scalars. Find a function $f: \mathbb{R} \rightarrow \mathbb{R}$ to fit the data

- $f(x) = b + wx$
- $f(x) = b + w_1x + w_2x^2$
- $f(x) = b + \sum_{i=1}^9 w_i x^i$

MSE training:

$$\min_w \frac{1}{N} \sum_{n=1}^N \|f(x^{(n)}) - y^{(n)}\|_2^2$$



Linear regression

- $f(\mathbf{x})$ is linear

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + \underbrace{b}_{\text{bias}}$$

b can be absorbed into a new vector θ and $f(\mathbf{x}) = \theta^\top \mathbf{x}$

where $\mathbf{w} \in R^m, b \in R$.

- Choose the cost function as the **mean squared error (MSE)**

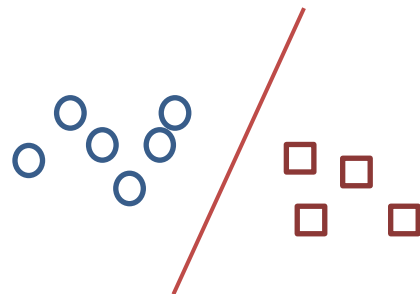
$$E = \frac{1}{2N} \sum_{n=1}^N (f(\mathbf{x}^{(n)}) - t^{(n)})^2 = \frac{1}{2N} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}^{(n)} + b - t^{(n)})^2$$

- Find optimal $\mathbf{w}^*$ and b^* by minimizing the cost function

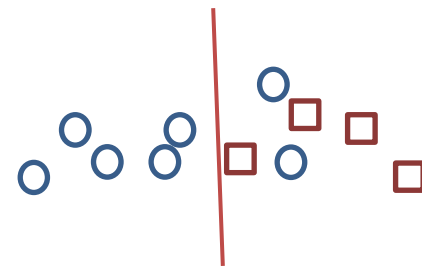
$$\left. \begin{aligned} \nabla_{\mathbf{w}} E &= \frac{1}{N} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}^{(n)} + b - t^{(n)}) \mathbf{x}^{(n)} = 0 \\ \nabla_b E &= \frac{1}{N} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}^{(n)} + b - t^{(n)}) = 0 \end{aligned} \right\} \mathbf{w}^*, b^*$$

Linear classification

- In the feature space, a linear classifier corresponds to a hyperplane



Linearly separable

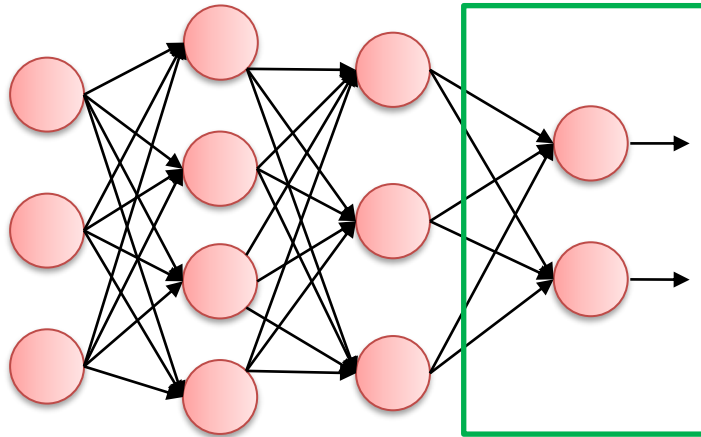


Linearly nonseparable

- Two typical linear classifiers
 - Perceptron
 - Support vector machine (SVM)

Discussion on classifiers

- What else classifiers do you know?
- If a neural network is used for classification, which classifier do you want to use in the **output layer**?

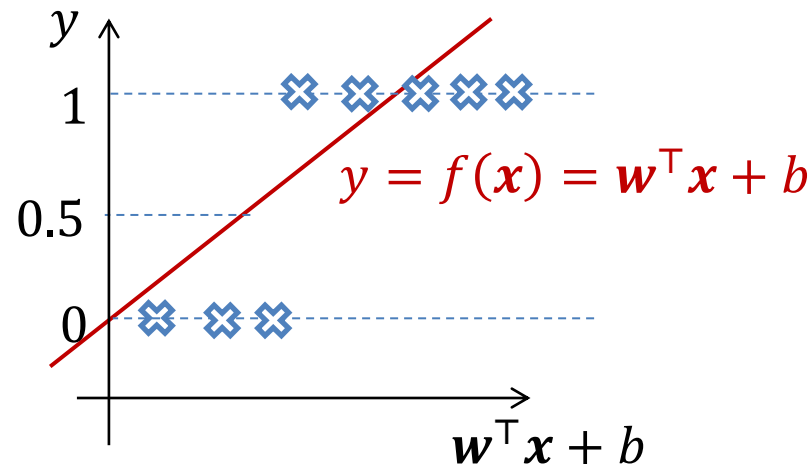


Any challenge here?

What shall we do?

Do binary classification using linear regression

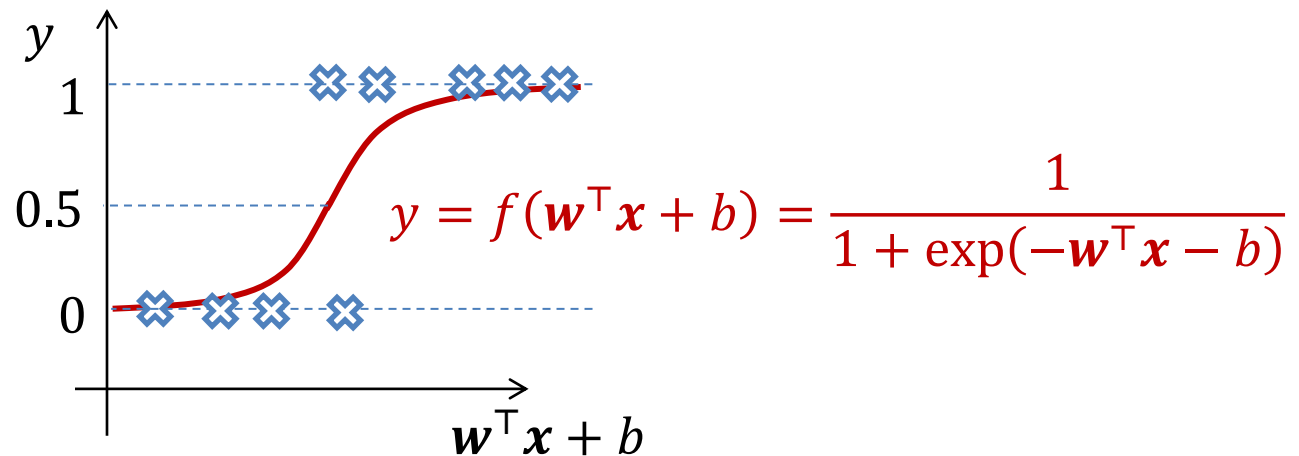
- Suppose $t \in \{0,1\}$. Consider $\mathbf{x} \in R^m$, can we use linear regression to do binary classification?



- Regression
 - Prediction $y = f(\mathbf{x})$ which is continuous
- Classification
 - Prediction $y = \begin{cases} 1, & \text{if } f(\mathbf{x}) \geq 0.5 \\ 0, & \text{if } f(\mathbf{x}) < 0.5 \end{cases}$

Do binary classification using nonlinear regression

- $f(x)$ can be a nonlinear functions, e.g., the **logistic sigmoid function**

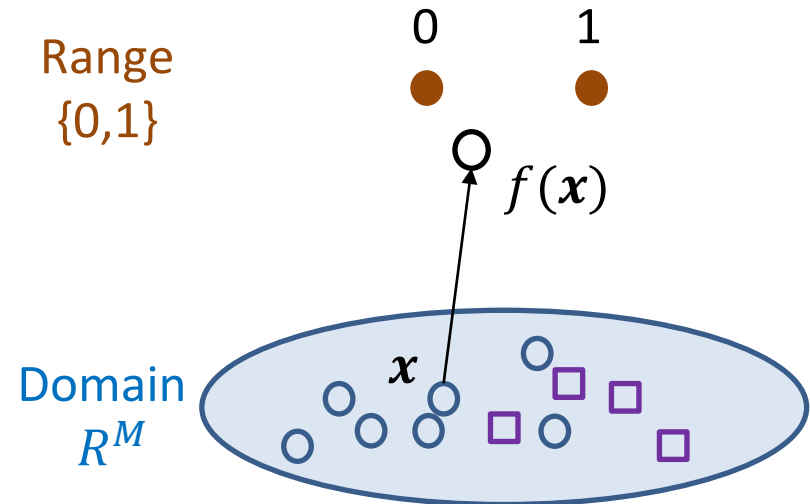
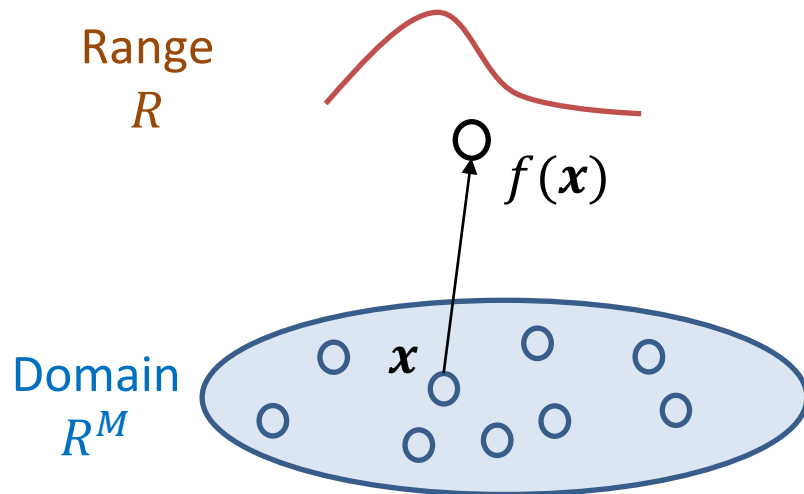


- Regression
 - Prediction $y = f(x)$ which is continuous
- Classification
 - Prediction $y = \begin{cases} 1, & \text{if } f(x) \geq 0.5 \\ 0, & \text{if } f(x) < 0.5 \end{cases}$

Input-output mapping

Regression

Classification



Train the nonlinear regression model

- $f(\mathbf{x})$ is a nonlinear function of $\mathbf{x}$
 - Usually it is a nonlinear function of $u = \mathbf{w}^\top \mathbf{x} + b$
- Choose the cost function as the **mean squared error (MSE)**

$$E = \frac{1}{2N} \sum_{n=1}^N (f(\mathbf{x}^{(n)}) - t^{(n)})^2$$

- Find optimal $\mathbf{w}^*$ and b^* by minimizing the cost function

$$\begin{aligned} \nabla_{\mathbf{w}} E &= \frac{1}{N} \sum_{n=1}^N (f(\mathbf{x}^{(n)}) - t^{(n)}) f'(u) \mathbf{x}^{(n)} = 0 \\ \nabla_b E &= \frac{1}{N} \sum_{n=1}^N (f(\mathbf{x}^{(n)}) - t^{(n)}) f'(u) = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} \nabla_{\mathbf{w}} E \\ \nabla_b E \end{aligned}} \right\} \mathbf{w}^*, b^*$$

Normal distribution assumption

- Assume the label follows a normal distribution with mean $f(\mathbf{x})$:

$$p(\underline{t}|\mathbf{x}) = \mathcal{N}(t; f(\mathbf{x})) \propto \exp\left(-\left(t - f(\mathbf{x})\right)^2\right)$$

(sometimes we may not distinguish between plain and italic type faces)

- Given a dataset $\{(\mathbf{x}^{(1)}, t^{(1)}), \dots, (\mathbf{x}^{(N)}, t^{(N)})\}$. View t and $\mathbf{x}$ as **random** variables. The **conditional data likelihood** function (independence assumption)

$$P(t^{(1)}, \dots, t^{(N)} | \mathbf{X}) = \prod_{n=1}^N P(t^{(n)} | \mathbf{x}^{(n)})$$

- $\max \log P(t^{(1)}, \dots, t^{(N)} | \mathbf{X})$ is equivalent to

$$\min E = \frac{1}{2N} \sum_{n=1}^N (f(\mathbf{x}^{(n)}) - t^{(n)})^2$$

Bernoulli distribution assumption for 2-class classification

$$p(t|\mathbf{x}) = \mathcal{N}(t; f(\mathbf{x})) \propto \exp\left(-\left(t - f(\mathbf{x})\right)^2\right)$$

- For regression (**t is continuous**), the normal distribution assumption is natural
- For classification (**t is discrete**), it is strange
- We have more suitable assumptions on the data distribution for classification
 - **Bernoulli distribution**

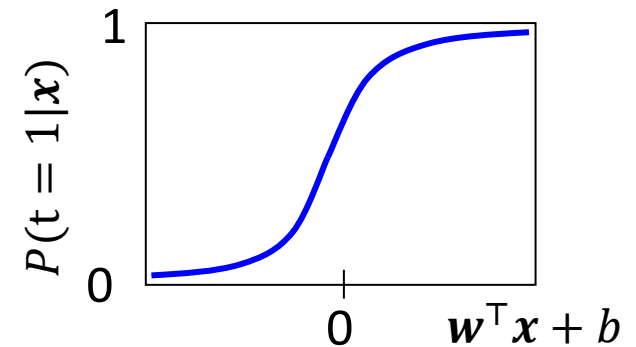
To do so, we need to first write down the Bernoulli distribution with (parameterized) probability $P(t = 1|\mathbf{x})$

Logistic regression

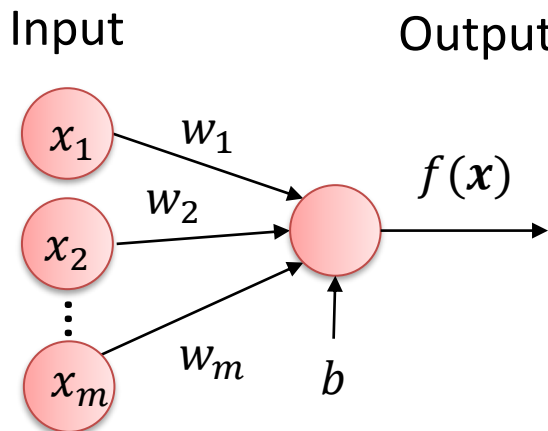
- Use the logistic sigmoid function to parameterize the Bernoulli distribution

$$P(t = 1|\mathbf{x}) = f(\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x} - b)}$$

$$P(t = 0|\mathbf{x}) = 1 - f(\mathbf{x})$$



- From the perspective of neural network



Label

$t^{(1)}$ $t^{(2)}$ $t^{(3)}$...

1 0 1 ...

For 2-class problems, one 0-1 unit is enough for representing a label

Maximum conditional data likelihood

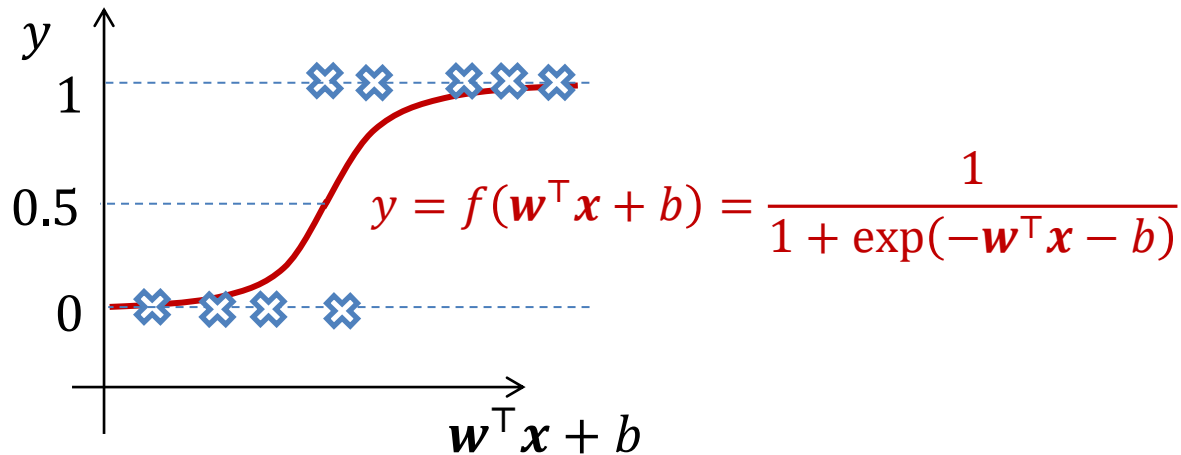
- *Recall* the maximum conditional likelihood estimation:
 - 1. write down the conditional likelihood function
 - 2. take log and maximize
- Given a dataset $\{(\mathbf{x}^{(1)}, t^{(1)}), \dots, (\mathbf{x}^{(N)}, t^{(N)})\}$ where $t^{(n)} \in \{0, 1\}$
- View t as a **Bernoulli** variable and $P(t = 1 | \mathbf{x}) = f(\mathbf{x}; \mathbf{w}, b)$. The conditional likelihood function

$$P(t^{(1)}, \dots, t^{(N)} | X; \mathbf{w}, b) = \prod_{n=1}^N f(\mathbf{x}^{(n)})^{t^{(n)}} (1 - f(\mathbf{x}^{(n)}))^{1-t^{(n)}}$$

- Maximizing the likelihood is equivalent to minimizing

$$\begin{aligned} E(\mathbf{w}, b) &= -\frac{1}{N} \ln P(t^{(1)}, \dots, t^{(N)}) && \text{Cross-entropy (CE) function} \\ &= -\frac{1}{N} \sum_{n=1}^N \left(t^{(n)} \ln f(\mathbf{x}^{(n)}) + (1 - t^{(n)}) \ln(1 - f(\mathbf{x}^{(n)})) \right) \end{aligned}$$

Sigmoid classifier with different loss functions



- Mean squared error (MSE) function

$$E(\mathbf{w}, b) = \frac{1}{2N} \sum_{n=1}^N (f(\mathbf{x}^{(n)}) - t^{(n)})^2$$

- Cross-entropy function

$$E(\mathbf{w}, b) = -\frac{1}{N} \sum_{n=1}^N \left(t^{(n)} \ln f(\mathbf{x}^{(n)}) + (1 - t^{(n)}) \ln(1 - f(\mathbf{x}^{(n)})) \right)$$

$f(\mathbf{x}^{(n)})$ is interpreted as a **probability**

Exercise: Calculate the gradient

Cross-entropy (CE) function

$$E(\mathbf{x}, b) = -\frac{1}{N} \sum_{n=1}^N \left(t^{(n)} \ln f(\mathbf{x}^{(n)}) + (1 - t^{(n)}) \ln(1 - f(\mathbf{x}^{(n)})) \right)$$

where $f(\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x} - b)}$

$$\nabla_{\mathbf{w}} E(\mathbf{w}, b) = ?$$

$$\nabla_b E(\mathbf{w}, b) = ?$$

$$h(z) = \frac{1}{1 + \exp(-z)}$$

$$h' = h(1 - h)$$

Training and testing

$$E(\mathbf{w}, b) = -\frac{1}{N} \sum_{n=1}^N \left(t^{(n)} \ln f(\mathbf{x}^{(n)}) + (1 - t^{(n)}) \ln(1 - f(\mathbf{x}^{(n)})) \right)$$

- Calculate the gradient

$$\nabla E(\mathbf{w}, b) = \frac{1}{N} \sum_n \mathbf{x}^{(n)} \left(f(\mathbf{x}^{(n)}) - t^{(n)} \right)$$

- Some regularization term can be incorporated into the cost function

$$J(\mathbf{w}, b) = E(\mathbf{w}, b) + \lambda ||\mathbf{w}||^2 / 2$$

- **Training:** learn $\boldsymbol{\theta}$ to minimize the cost function

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} J(\mathbf{w}, b)$$

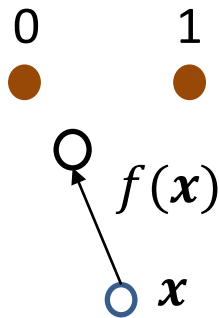
$$b \leftarrow b - \eta \nabla_b J(\mathbf{w}, b)$$

where η is the learning rate

- **Testing:** for a new input $\mathbf{x}$, if $P(t = 1|\mathbf{x}) > P(t = 0|\mathbf{x})$ then we predict the input as class 1, and 0 otherwise

Summary of part 1

Function mapping



Linear
regression



nonlinear
regression

$$f(x) = \mathbf{w}^\top \mathbf{x} + b$$

$$f(x) = g(\mathbf{w}^\top \mathbf{x} + b)$$

MLE

1. normal distribution:

- $g(\cdot)$ is any nonlinear function
- Loss function: MSE

2. Bernoulli distribution:

- $g(\cdot)$ is sigmoid function
- Loss function: Cross-entropy

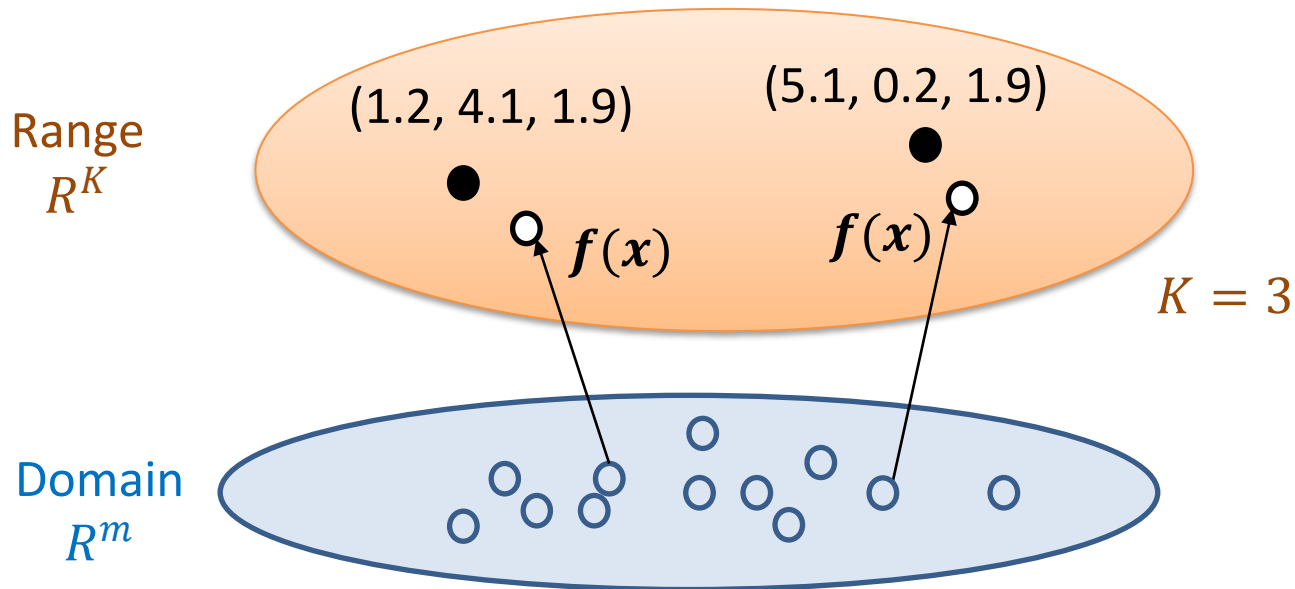
Logistic regression!

Outline

- I. Logistic regression
- II. Softmax regression
- III. Summary

Linear regression for vectors

- If $t \in R^K$ is a continuous vector, then use a **linear** function $f_k(\mathbf{x})$ to regress t_k for $k = 1, \dots, K$: $f_k(\mathbf{x}) = \mathbf{w}_k^\top \mathbf{x} + b_k$, where $\mathbf{w}_k \in R^m, b_k \in R$



Linear regression for vectors

- $f_k(\mathbf{x})$ is linear for $k = 1, \dots, K$: $f_k(\mathbf{x}) = \mathbf{w}_k^\top \mathbf{x} + b_k$, where $\mathbf{w}_k \in R^m, b_k \in R$.
- Choose the cost function as the **mean squared error (MSE)**

$$E = \frac{1}{2N} \sum_{n=1}^N \sum_{k=1}^K \left(f_k(\mathbf{x}^{(n)}) - t_k^{(n)} \right)^2$$

- Find optimal $\mathbf{w}_k^*$ and b_k^* by solving the **linear system**

$$\begin{aligned} \nabla_{\mathbf{w}_k} E &= \frac{1}{N} \sum_{n=1}^N \left(\mathbf{w}_k^\top \mathbf{x}^{(n)} + b_k - t_k^{(n)} \right) \mathbf{x}^{(n)} = 0 \\ \nabla_{b_k} E &= \frac{1}{N} \sum_{n=1}^N \left(\mathbf{w}_k^\top \mathbf{x}^{(n)} + b_k - t_k^{(n)} \right) = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} \nabla_{\mathbf{w}_k} E \\ \nabla_{b_k} E \end{aligned}} \right\} \mathbf{w}_k^*, b_k^*$$

Nonlinear regression for vectors

- $f_k(\mathbf{x})$ is a **nonlinear** function of $u_k = \mathbf{w}_k^\top \mathbf{x} + b_k$, where $\mathbf{w}_k \in R^m, b_k \in R$.

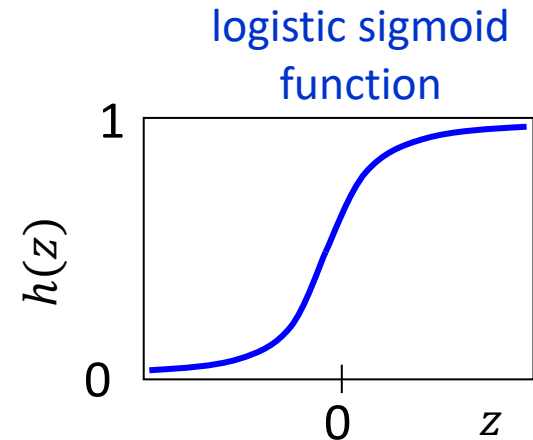
– E.g. $f_k(\mathbf{x}) = h(\mathbf{w}_k^\top \mathbf{x} + b_k)$

$$h(z) = \frac{1}{1 + \exp(-z)} \quad \text{Or other functions}$$

- Choose the cost function as the **MSE**

$$E = \frac{1}{N} \sum_{n=1}^N E^{(n)}$$

$$\text{where } E^{(n)} = \frac{1}{2} \sum_{k=1}^K \left(f_k(\mathbf{x}^{(n)}) - t_k^{(n)} \right)^2$$



Gradient of the cost function

$$E = \frac{1}{N} \sum_{n=1}^N E^{(n)} \quad \text{where } E^{(n)} = \frac{1}{2} \sum_{k=1}^K \left(f_k(\mathbf{x}^{(n)}) - t_k^{(n)} \right)^2$$

- Denote $u_k^{(n)} = \mathbf{w}_k^\top \mathbf{x}^{(n)} + b_k$ and $\delta_k^{(n)} = \frac{\partial E^{(n)}}{\partial u_k^{(n)}}$, then

$$\frac{\partial E^{(n)}}{\partial \mathbf{w}_k} = \delta_k^{(n)} \mathbf{x}^{(n)}$$

$$\frac{\partial E^{(n)}}{\partial b_k} = \delta_k^{(n)} \frac{\partial u_k^{(n)}}{\partial b_k} = \delta_k^{(n)}$$

$\delta_k^{(n)}$: Local sensitivity or local gradient

$$\delta_k^{(n)} = \left(f(u_k^{(n)}) - t_k \right) f'(u_k^{(n)})$$

Vector-matrix form

- Define

$$\mathbf{W} = \begin{pmatrix} w_{11} & \cdots & w_{1m} \\ \vdots & \vdots & \vdots \\ w_{K1} & \cdots & w_{Km} \end{pmatrix} \quad \frac{\partial E}{\partial \mathbf{W}} = \begin{pmatrix} \partial E / \partial w_{11} & \cdots & \partial E / \partial w_{1m} \\ \vdots & \vdots & \vdots \\ \partial E / \partial w_{K1} & \cdots & \partial E / \partial w_{Km} \end{pmatrix}$$

m : the number of inputs; K : the number of outputs

- Output: $\mathbf{f}(\mathbf{x})$, where $\mathbf{f}, \mathbf{b} \in R^K$, $\mathbf{x} \in R^m$

- Error function: $E = \frac{1}{2N} \sum_{n=1}^N \left\| \mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)} \right\|_2^2$

- Gradient

Elementwise product

$$\nabla_{\mathbf{W}} E = \frac{1}{N} \sum_{n=1}^N (\mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)}) \odot \mathbf{f}'(\mathbf{u}^{(n)}) (\mathbf{x}^{(n)})^\top \in R^{K \times m}$$

$$\nabla_{\mathbf{b}} E = \frac{1}{N} \sum_{n=1}^N (\mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)}) \odot \mathbf{f}'(\mathbf{u}^{(n)}) \in R^K$$

Representation of class labels

- For classification, given $\{(\mathbf{x}^{(1)}, t^{(1)}), \dots, (\mathbf{x}^{(N)}, t^{(N)})\}$, the goal is to find a mapping from $\mathbf{x}^{(n)}$ to $t^{(n)}$

$$f: \mathbb{R}^m \rightarrow \Omega$$

where Ω is a discrete set

- $t^{(n)}$ can be a (discrete) scalar or vector

Suppose there are 5 classes in total

Rarely
used

Scalar representation

$$t^{(1)} = 1$$

$$t^{(3)} = 3$$

Vector representation

$$\mathbf{t}^{(1)} = (1, 0, 0, 0, 0)^\top$$

$$\mathbf{t}^{(3)} = (0, 0, 1, 0, 0)^\top$$

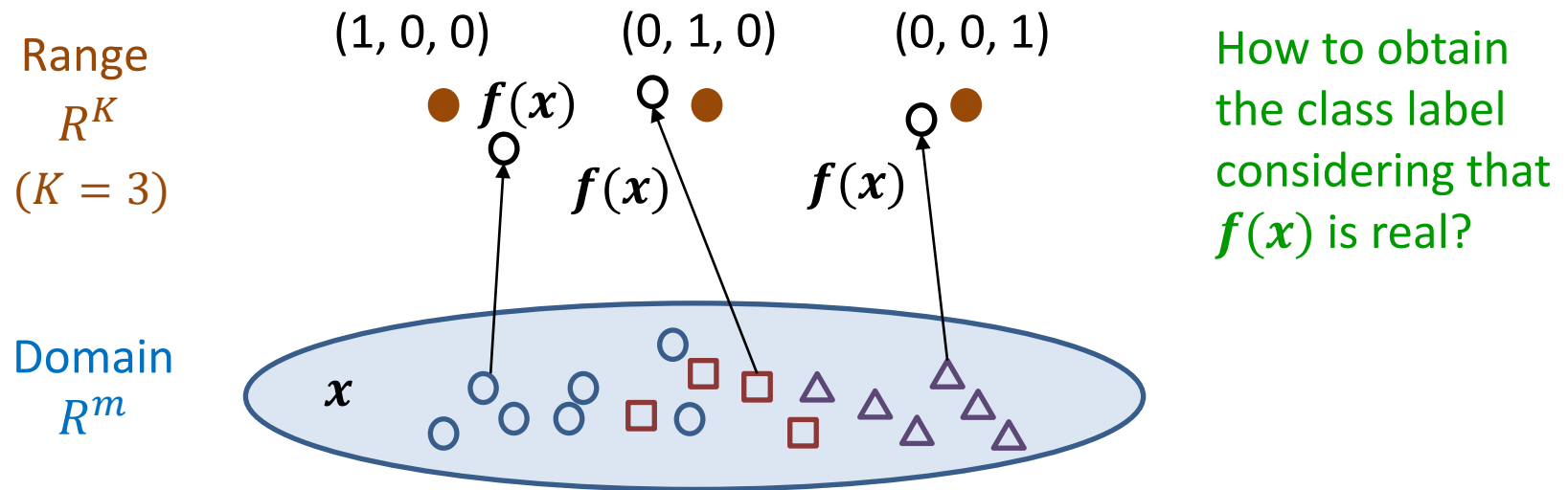
Usually
used

Any problem with
scalar
representation?

- 1-of-K representation
- Property: $t_k^{(n)} \in \{0, 1\}$; $\sum_k t_k^{(n)} = 1$

Do multilabel classification using regression

- Using the 1-of-K representation for class labels, one can also do classification using linear regression



- Both linear and nonlinear regression discussed before can be applied

Normal distribution assumption

- Assume the label follows a normal distribution with mean $\mathbf{f}(\mathbf{x})$:

$$p(\mathbf{t}|\mathbf{x}) = \mathcal{N}(\mathbf{t}; \mathbf{f}(\mathbf{x}), \mathbf{I}) \propto \exp\left(-\|\mathbf{t} - \mathbf{f}(\mathbf{x})\|_2^2\right)$$

($\mathbf{f}(\mathbf{x})$ determines the prob of the sample's label, therefore it's the output)

- Given a dataset $\{(\mathbf{x}^{(1)}, \mathbf{t}^{(1)}), \dots, (\mathbf{x}^{(N)}, \mathbf{t}^{(N)})\}$. View $\mathbf{t}$ and $\mathbf{x}$ as **random** variables. The **conditional data likelihood** function (independence assumption)

$$P(\mathbf{t}^{(1)}, \dots, \mathbf{t}^{(N)} | \mathbf{X}) = \prod_{n=1}^N P(\mathbf{t}^{(n)} | \mathbf{x}^{(n)})$$

- $\max \log P(\mathbf{t}^{(1)}, \dots, \mathbf{t}^{(N)} | \mathbf{X})$ is equivalent to

$$\min E = \frac{1}{2N} \sum_{n=1}^N \|\mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)}\|_2^2$$

Multinoulli distribution assumption for classification

$$p(\mathbf{t}|\mathbf{x}) = \mathcal{N}(\mathbf{t}; \mathbf{f}(\mathbf{x}), \mathbf{I}) \propto \exp\left(-\|\mathbf{t} - \mathbf{f}(\mathbf{x})\|_2^2\right)$$

- For regression (**t is continuous**), the normal distribution assumption is natural
- For classification (**t is discrete**), it is strange
- We have more reasonable assumptions on the data distribution for classification
 - Bernoulli distribution for $K = 2$
 - Multinoulli or categorical distribution for $K > 2$

To do so, we need to first write down the multinoulli distribution with (parameterized) probability $P(t_k = 1|\mathbf{x})$

Recap: multinoulli or categorical prob distributions

- The prob distribution over a single discrete variable t with K different states where K is finite

$$P(t = k|\mathbf{p}) = p_k$$

where $\mathbf{p} \in [0,1]^K$ and $\sum_{k=1}^K p_k = 1$

- With one-hot representation $\mathbf{t} = (0, \dots, 1, \dots, 0)^\top$, then

$$P(\mathbf{t}) = \prod_{k=1}^K P(t_k = 1)^{t_k}$$

A random variable

A value 0 or 1

Scalar: 1,2,3

$$P(t = 1) = 0.2;$$

$$P(t = 2) = 0.5;$$

$$P(t = 3) = 0.3$$

One-hot: $(1,0,0)^\top, (0,1,0)^\top, (0,0,1)^\top$

$$P(\mathbf{t} = (1,0,0)^\top)$$

$$= P(t_1 = 1)^1 P(t_2 = 1)^0 P(t_3 = 1)^0 = 0.2$$

Similarly, $P(\mathbf{t} = (0,1,0)^\top) = 0.5;$

$$P(\mathbf{t} = (0,0,1)^\top) = 0.3$$

Formulation

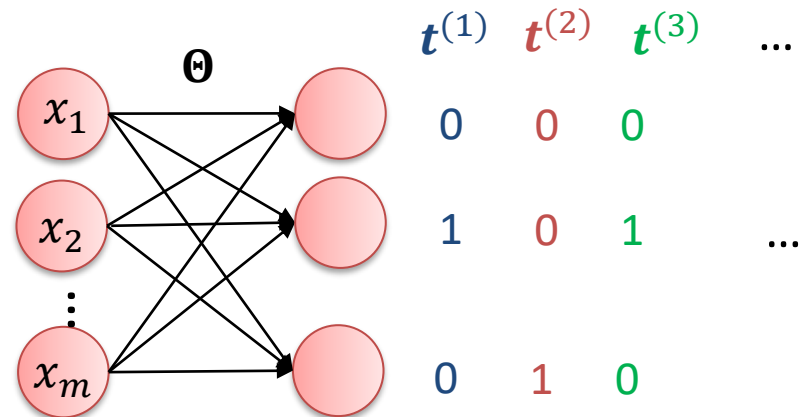
- For K -class problems ($K > 2$), we try to learn a Multinoulli distribution $P(\mathbf{t} = k|\mathbf{x})$ where $k = 1, \dots, K$

- With one-hot representation $\mathbf{t} = (0, \dots, 1, \dots, 0)^\top$, then

$$P(\mathbf{t}|\mathbf{x}) = \prod_{k=1}^K P(\mathbf{t}_k = 1|\mathbf{x})^{t_k}$$

A random variable

A value



- Let $P(\mathbf{t}_k = 1|\mathbf{x})$ take the following form

$$P(\mathbf{t}_k = 1|\mathbf{x}) = \frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x})} \triangleq f_k(\mathbf{x})$$

Clearly, $f_k(\mathbf{x}) \in (0,1)$ and $\sum_{k=1}^K f_k(\mathbf{x}) = 1$

$f(\mathbf{x})$ determines the prob of the sample's label, therefore it's the output

Softmax function

$$f_k(\mathbf{x}) = \frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x})}$$

- The following function is called *softmax* function

$$\psi(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} = \frac{\exp(z_i)}{\exp(z_i) + \sum_{j \neq i} \exp(z_j)} \in (0, 1)$$

- If $z_i > z_j$ for all $j \neq i$
 - Then $\psi(z_i) > \psi(z_j)$ for all $j \neq i$ but it is smaller than 1
- If $z_i \gg z_j$ for all $j \neq i$,
 - then $\psi(z_i) \rightarrow 1$ and $\psi(z_j) \rightarrow 0$ for $j \neq i$.

Maximum conditional likelihood

- Since

$$P(\mathbf{t}|\mathbf{p}) = \prod_{k=1}^K P(t_k = 1)^{t_k}$$

- Given a dataset $\{(\mathbf{x}^{(1)}, \mathbf{t}^{(1)}), \dots, (\mathbf{x}^{(N)}, \mathbf{t}^{(N)})\}$. The conditional likelihood function (independence assumption):

$$P(\mathbf{t}^{(1)}, \dots, \mathbf{t}^{(N)} | \mathbf{X}) = \prod_{n=1}^N \prod_{k=1}^K P(t_k^{(n)} = 1 | \mathbf{x}^{(n)})^{t_k^{(n)}}$$

- Maximizing this likelihood function is equivalent to minimizing

$$E(\Theta) = -\frac{1}{N} \ln P(\mathbf{t}^{(1)}, \dots, \mathbf{t}^{(N)})$$

Take log and negative,
then minimize

$$= -\frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K t_k^{(n)} \ln \left[\frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x}^{(n)})} \right] \leftarrow f_k(\mathbf{x}^{(n)}) \text{ or } f_k^{(n)}$$

Cross-entropy
function

$$= -\frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K t_k^{(n)} \ln f_k^{(n)} = -\frac{1}{N} \sum_{n=1}^N \ln P(t_k^{(n)} = 1 | \mathbf{x}^{(n)})$$

Cross-entropy (CE) function

$$\begin{aligned} E(\boldsymbol{\Theta}) &= -\frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K t_k^{(n)} \ln \frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x}^{(n)})} \\ &= -\frac{1}{N} \sum_{n=1}^N \ln P(t_k^{(n)} = 1 | \mathbf{x}^{(n)}) \end{aligned}$$

where $\boldsymbol{\Theta} = \begin{bmatrix} \boldsymbol{\theta}^{(1)} & \boldsymbol{\theta}^{(2)} & \dots & \boldsymbol{\theta}^{(K)} \end{bmatrix}^\top$.

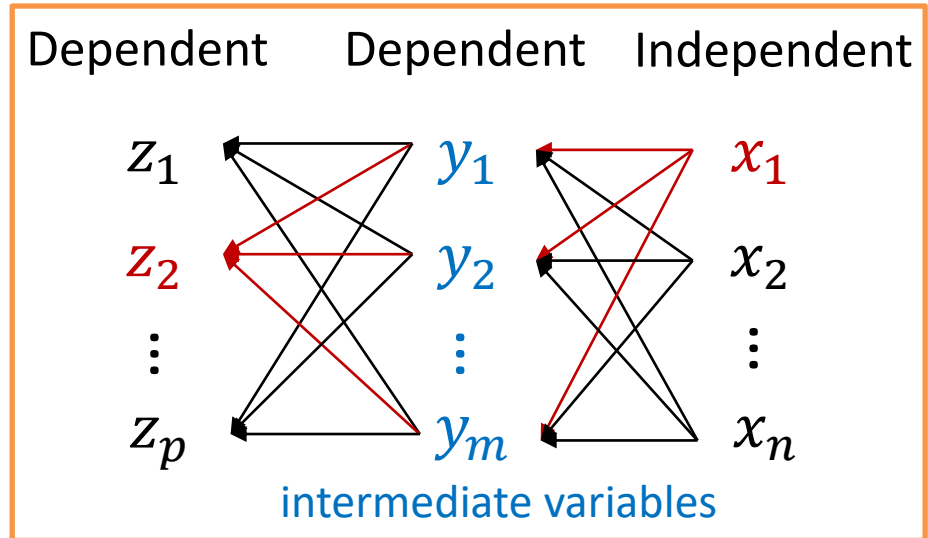
Why is it called “cross-entropy” function?



Watch the supplementary video
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Recap: Derivative of two-step composition

- Independent variables
 $x_1, x_2, \dots, x_n$
- Each y_i is a function of
 $x_1, x_2, \dots, x_n$
- Each z_i is a function of
 $y_1, y_2, \dots, y_m$



What's partial derivative of z_i w.r.t. x_j ?

$$\frac{\partial z_i}{\partial x_j} = \sum_{k=1}^m \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

Sum over the intermediate variables

for any $i \in \{1, 2, \dots, p\}$ and $j \in \{1, 2, \dots, n\}$

Calculate the gradient

$$E(\boldsymbol{\theta}) = \frac{1}{N} \sum_{n=1}^N E^{(n)}(\boldsymbol{\theta}), \quad E^{(n)}(\boldsymbol{\theta}) = - \sum_{i=1}^K t_i^{(n)} \ln f_i(\mathbf{x}^{(n)})$$

where

$$f_i(\mathbf{x}^{(n)}) = \frac{\exp(u_i^{(n)})}{\sum_{j=1}^K \exp(u_j^{(n)})}, \quad u_k^{(n)} = \boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)}$$

$$\frac{\partial E^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} = \frac{\partial E^{(n)}}{\partial u_k^{(n)}} \frac{\partial u_k^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} = \sum_{i=1}^K \frac{\partial E^{(n)}}{\partial f_i^{(n)}} \frac{\partial f_i^{(n)}}{\partial u_k^{(n)}} \frac{\partial u_k^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} \quad \left[f_i^{(n)} \triangleq f_i(\mathbf{x}^{(n)}) \right]$$

Local sensitivity or
local gradient

$$\frac{\partial E^{(n)}}{\partial f_i^{(n)}} = -t_i^{(n)} \frac{1}{f_i^{(n)}}$$

$$?$$

$$\frac{\partial u_k^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} = \mathbf{x}^{(n)}$$

Calculate the gradient

$$E(\boldsymbol{\Theta}) = \frac{1}{N} \sum_{n=1}^N E^{(n)}(\boldsymbol{\Theta}), \quad E^{(n)}(\boldsymbol{\Theta}) = - \sum_{i=1}^K t_i^{(n)} \ln f_i(\mathbf{x}^{(n)})$$

where

$$f_i(\mathbf{x}^{(n)}) = \frac{\exp(u_i^{(n)})}{\sum_{j=1}^K \exp(u_j^{(n)})}, \quad u_k^{(n)} = \boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)}$$

For u_k that appears only in the denominator ($k \neq i$)

$$\frac{\partial f_i^{(n)}}{\partial u_k^{(n)}} = - \frac{\exp(u_i^{(n)}) \exp(u_k^{(n)})}{\left(\sum_j \exp(u_j^{(n)}) \right)^2} = -f_i^{(n)} f_k^{(n)}$$

For u_i , it appears in both numerator and denominator

$$\frac{\partial f_i^{(n)}}{\partial u_i^{(n)}} = \frac{\exp(u_i^{(n)})}{\sum_j \exp(u_j^{(n)})} - \frac{\left(\exp(u_i^{(n)}) \right)^2}{\left(\sum_j \exp(u_j^{(n)}) \right)^2} = f_i^{(n)} (1 - f_i^{(n)})$$

Therefore

$$\frac{\partial f_i^{(n)}}{\partial u_k^{(n)}} = f_i^{(n)} (\Delta_{i,k} - f_k^{(n)}) \quad \text{where} \quad \Delta_{i,k} = \begin{cases} 1, & \text{if } i = k \\ 0, & \text{else.} \end{cases}$$

Calculate the gradient

$$E(\boldsymbol{\Theta}) = \frac{1}{N} \sum_{n=1}^N E^{(n)}(\boldsymbol{\Theta}), \quad E^{(n)}(\boldsymbol{\Theta}) = - \sum_{i=1}^K t_i^{(n)} \ln f_i(\mathbf{x}^{(n)})$$

where

$$f_i(\mathbf{x}^{(n)}) = \frac{\exp(u_i^{(n)})}{\sum_{j=1}^K \exp(u_j^{(n)})}, \quad u_k^{(n)} = \boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)}$$

$$\begin{aligned} \frac{\partial E^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} &= \sum_{i=1}^K \frac{\partial E^{(n)}}{\partial f_i^{(n)}} \frac{\partial f_i^{(n)}}{\partial u_k^{(n)}} \frac{\partial u_k^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} \\ &= \sum_{i=1}^K \left(-t_i^{(n)} \frac{1}{f_i^{(n)}} \right) \left(f_i^{(n)} (\Delta_{i,k} - f_k^{(n)}) \right) \left(\mathbf{x}^{(n)} \right) \\ &= - \left(\sum_{i=1}^K t_i^{(n)} \Delta_{i,k} - \underbrace{\sum_{i=1}^K t_i^{(n)} f_k^{(n)}}_{=1} \right) \mathbf{x}^{(n)} \\ &= - \left(t_k^{(n)} - f_k^{(n)} \right) \mathbf{x}^{(n)} \end{aligned}$$

Calculate the gradient

$$E(\boldsymbol{\theta}) = \frac{1}{N} \sum_{n=1}^N E^{(n)}(\boldsymbol{\theta}), \quad E^{(n)}(\boldsymbol{\theta}) = - \sum_{i=1}^K t_i^{(n)} \ln f_i(\mathbf{x}^{(n)})$$

where $f_i(\mathbf{x}^{(n)}) = \frac{\exp(u_i^{(n)})}{\sum_{j=1}^K \exp(u_j^{(n)})}$, $u_k^{(n)} = \boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)}$

$$\frac{\partial E^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} = \delta_k^{(n)} \mathbf{x}^{(n)}, \text{ where } \delta_k^{(n)} \triangleq \frac{\partial E^{(n)}}{\partial u_k^{(n)}} = - \left(t_k^{(n)} - f_k^{(n)} \right)$$

is the **local gradient or local sensitivity**.

Average over samples

$$\begin{aligned} \frac{\partial E}{\partial \boldsymbol{\theta}^{(k)}} &= \frac{1}{N} \sum_{n=1}^N \frac{\partial E^{(n)}}{\partial \boldsymbol{\theta}^{(k)}} = - \frac{1}{N} \sum_{n=1}^N \left(t_k^{(n)} - f_k^{(n)} \right) \mathbf{x}^{(n)} \\ &= - \frac{1}{N} \sum_{n=1}^N \left(t_k^{(n)} - P(t_k^{(n)} = 1 | \mathbf{x}^{(n)}) \right) \mathbf{x}^{(n)} \end{aligned}$$

Vector-matrix form

- Note the definitions

$$\Theta = \begin{pmatrix} \theta_{11} & \cdots & \theta_{1m} \\ \vdots & \vdots & \vdots \\ \theta_{K1} & \cdots & \theta_{Km} \end{pmatrix} \quad \frac{\partial E}{\partial \Theta} = \begin{pmatrix} \partial E / \partial \theta_{11} & \cdots & \partial E / \partial \theta_{1m} \\ \vdots & \vdots & \vdots \\ \partial E / \partial \theta_{K1} & \cdots & \partial E / \partial \theta_{Km} \end{pmatrix}$$

m : the number of inputs; K : the number of outputs

- Output: $\mathbf{f}(\mathbf{x})$ is the softmax function where $\mathbf{f} \in R^K$, $\mathbf{x} \in R^m$
- The gradient of the cross-entropy error function

$$\nabla_{\Theta} E = \begin{pmatrix} (\partial E / \partial \Theta^{(1)})^{\top} \\ \vdots \\ (\partial E / \partial \Theta^{(K)})^{\top} \end{pmatrix} = \frac{1}{N} \sum_{n=1}^N (\mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)}) (\mathbf{x}^{(n)})^{\top} \in R^{K \times m}$$

Training and testing

- Calculate the gradient of the cross-entropy error function

$$\nabla_{\Theta} E = \frac{1}{N} \sum_{n=1}^N \left(f(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)} \right) (\mathbf{x}^{(n)})^{\top}$$

- As before, some regularization term can be incorporated into the cost function

$$J(\Theta) = E(\Theta) + \lambda ||\Theta||^2 / 2$$

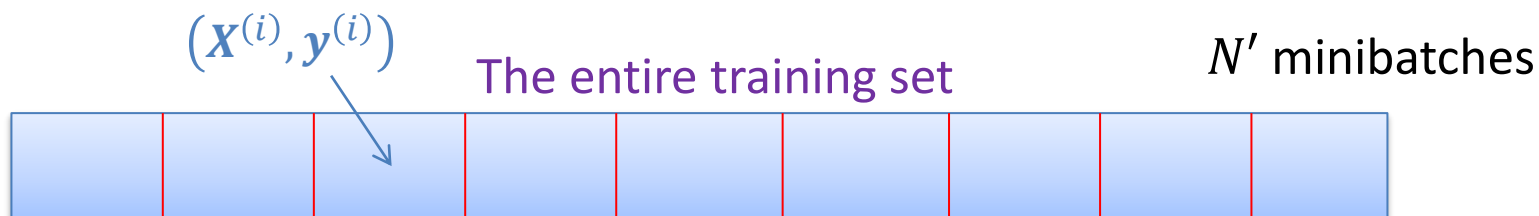
- **Training:** minimize the cost function with gradient $\nabla J(\Theta)$

$$\Theta \leftarrow \Theta - \alpha \nabla J(\Theta)$$

where α is the learning rate

- **Testing:** find the maximum $P(t_k = 1 | \mathbf{x})$ among k for a new input $\mathbf{x}$

Recall: Stochastic gradient decent (SGD)



- Minimizing the cost function over the entire training set is computationally expensive
- Decompose the training set into **minibatches** and optimize the cost function $L(\mathbf{X}^{(i)}, \mathbf{y}^{(i)}; \boldsymbol{\theta})$ defined over individual minibatches $(\mathbf{X}^{(i)}, \mathbf{y}^{(i)})$

- $J(\boldsymbol{\theta}) = \sum_{i=1}^{N'} L(\mathbf{X}^{(i)}, \mathbf{y}^{(i)}; \boldsymbol{\theta})$
- The batchsize ranges from 1 to a few hundreds

- At every iteration, update $\boldsymbol{\theta}$ as follows

$$\begin{aligned}\boldsymbol{\theta} &= \boldsymbol{\theta} - \eta \mathbf{g}' \\ \mathbf{g}' &= \nabla_{\boldsymbol{\theta}} L(\mathbf{X}^{(i)}, \mathbf{y}^{(i)}; \boldsymbol{\theta})\end{aligned}$$

Introducing bias

- So far we have assumed

$$f_k(\mathbf{x}) = P(t_k = 1|\mathbf{x}) = \frac{\exp(u_k^{(n)})}{\sum_{j=1}^K \exp(u_j^{(n)})} \quad u_k^{(n)} = \boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)}$$

- Sometimes a bias is introduced into $u_k^{(n)}$ and the parameters become $\{\mathbf{W}, \mathbf{b}\}$

$$u_k^{(n)} = \mathbf{w}^{(k)\top} \mathbf{x}^{(n)} + b^{(k)}$$

- It's easy to show that
$$\frac{\partial E}{\partial \mathbf{w}^{(k)}} = -\frac{1}{N} \sum_{n=1}^N \left(t_k^{(n)} - f_k(\mathbf{x}^{(n)}) \right) \mathbf{x}^{(n)}$$
$$\frac{\partial E}{\partial b^{(k)}} = -\frac{1}{N} \sum_{n=1}^N \left(t_k^{(n)} - f_k(\mathbf{x}^{(n)}) \right)$$

- Regularization is often applied on $\mathbf{W}$ only **Why?**

$$J(\mathbf{W}, \mathbf{b}) = E(\mathbf{W}, \mathbf{b}) + \lambda ||\mathbf{W}||^2 / 2$$

Softmax is over-parameterized

- The hypothesis

$$f_k(\mathbf{x}) = \frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x})} = \frac{\exp((\boldsymbol{\theta}^{(k)} - \boldsymbol{\phi})^\top \mathbf{x})}{\sum_{j=1}^K \exp((\boldsymbol{\theta}^{(j)} - \boldsymbol{\phi})^\top \mathbf{x})}$$

Then the new parameters $\hat{\boldsymbol{\theta}}^{(k)} \equiv \boldsymbol{\theta}^{(k)} - \boldsymbol{\phi}$ will result in the same prediction

- Minimizing the cross-entropy function has infinite number of solutions since

$$E(\boldsymbol{\theta}) = - \sum_{n=1}^N \sum_{k=1}^K t_k^{(n)} \ln \frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x}^{(n)})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x}^{(n)})} = E(\boldsymbol{\theta} - \boldsymbol{\Phi})$$

where $\boldsymbol{\Phi} = (\boldsymbol{\phi}, \dots, \boldsymbol{\phi})$

Relationship between logistic regression and softmax regression

Logistic
regression:

$$P(t = 1|\mathbf{x}) = \frac{1}{1 + \exp(-\boldsymbol{\theta}^\top \mathbf{x})}$$
$$P(t = 0|\mathbf{x}) = 1 - P(t = 1|\mathbf{x})$$

Softmax
regression:

$$P(t_k = 1|\mathbf{x}) = \frac{\exp(\boldsymbol{\theta}^{(k)\top} \mathbf{x})}{\sum_{j=1}^K \exp(\boldsymbol{\theta}^{(j)\top} \mathbf{x})}$$

Is there any relationship between them?

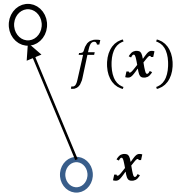


Watch the supplementary video
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Summary of part 2

Function mapping

(0,0,1) (0,1,0) (0,1,0)



Linear
regression



nonlinear
regression

$$f(x) = W^T x + b$$

$$f(x) = g(W^T x + b)$$

MLE

1. normal distribution:
 - $g(\cdot)$ is any nonlinear function
 - Loss function: MSE
2. multinoulli distribution:
 - $g(\cdot)$ is the softmax function
 - Loss function: Cross-entropy

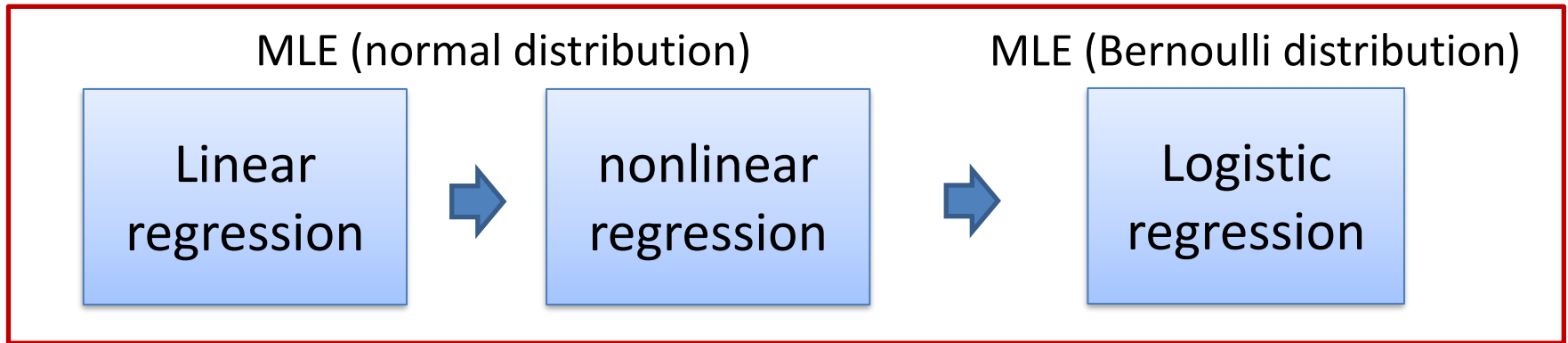
Softmax regression!

Outline

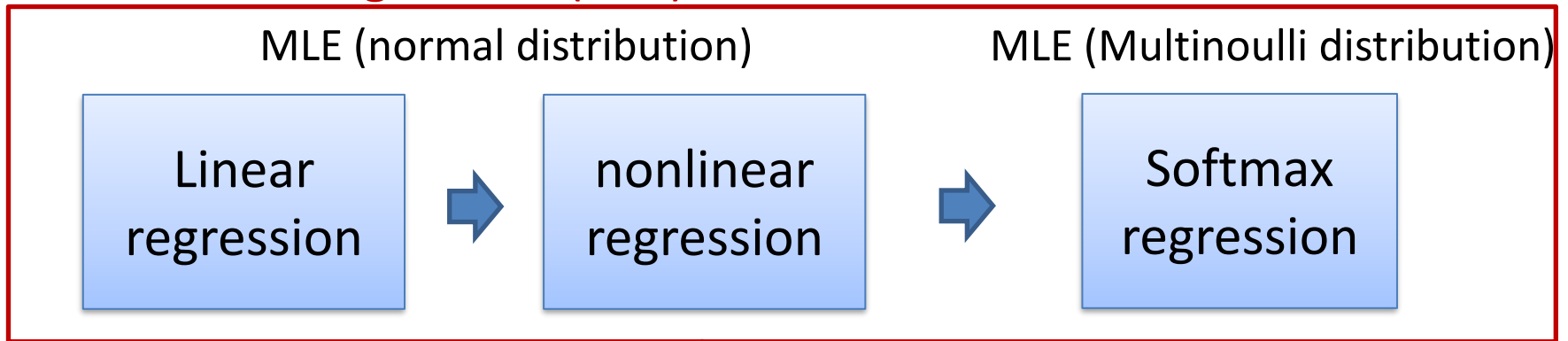
- I. Logistic regression
- II. Softmax regression
- III. Summary

Summary of this lecture

I. Logistic regression ($K=2$)



II. Softmax regression ($K>2$)



↑
If the task is regression,
we stop here

Summary of this lecture

- Nonlinear regression (linear regression as a special case)
 - Output: $\mathbf{f}(\mathbf{x}) = \mathbf{h}(\mathbf{\Theta}\mathbf{x})$, where $\mathbf{h}$ could be any nonlinear function
 - MSE: $E = \frac{1}{N} \sum_{n=1}^N E^{(n)}$, $E^{(n)} = \frac{1}{2} \left\| \mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)} \right\|_2^2$
 - Gradient: $\nabla_{\mathbf{\Theta}} E = \frac{1}{N} \sum_{n=1}^N (\mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)}) \odot \mathbf{f}'(\mathbf{u}^{(n)}) (\mathbf{x}^{(n)})^\top$
- Softmax regression (logistic regression as a special case)
 - Output: $\mathbf{f}(\mathbf{x}) = \mathbf{h}(\mathbf{\Theta}\mathbf{x})$, where $\mathbf{h}$ is the softmax function
 - Cross-entropy error: $E = \frac{1}{N} \sum_{n=1}^N E^{(n)}$, $E^{(n)} = -(\mathbf{t}^{(n)})^\top \ln \mathbf{f}^{(n)}$
 - Gradient: $\nabla_{\mathbf{\Theta}} E = \frac{1}{N} \sum_{n=1}^N (\mathbf{f}(\mathbf{x}^{(n)}) - \mathbf{t}^{(n)}) (\mathbf{x}^{(n)})^\top$